

Multiple Choice (10 marks)

1. A three-dimensional harmonic oscillator is in thermal equilibrium with a temperature reservoir at temperature T . The average total energy of the oscillator is
 - (a) $(1/2) k T$
 - (b) kT
 - (c) $(3/2) k T$
 - (d) $3 kT$
2. The speed of light inside of a nonmagnetic dielectric material with a dielectric constant of 4.0 is
 - (a) 1.2×10^9 m/s
 - (b) 3.0×10^8 m/s
 - (c) 1.5×10^8 m/s
 - (d) 1.0×10^8 m/s
3. The quantum efficiency of a photon detector is 0.1. If 100 photons are sent into the detector, one after the other, the detector will detect photons
 - (a) exactly 10 times
 - (b) an average of 10 times, with an rms deviation of about 0.1
 - (c) an average of 10 times, with an rms deviation of about 1
 - (d) an average of 10 times, with an rms deviation of about 3
4. Which of the following ions CANNOT be used as a dopant in germanium to make an n-type semiconductor?
 - (a) As
 - (b) P
 - (c) Sb
 - (d) B
5. For which of the following thermodynamic processes is the increase in the internal energy of an ideal gas equal to the heat added to the gas?
 - (a) Constant temperature
 - (b) Constant volume
 - (c) Constant pressure
 - (d) Adiabatic

True / False (10 marks)

Answer True or False

6. True or False: The electric displacement current through a surface S is proportional to the rate of change of the electric flux through S .
7. True or False: The electric displacement current through a surface S is proportional to the rate of change of the electric flux through S .
8. True or False: The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately 1.0 GeV/c².
9. True or False: If the voltage across a resistor is doubled, the rate of energy dissipation increases to 4 W due to the square of the voltage relationship.
10. True or False: If the total energy of a particle is twice its rest energy, its relativistic momentum is mc .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the impact of closing one end of an organ pipe on the harmonics produced. Which of the initial harmonics will remain, and why? Provide a detailed analysis.
12. Discuss how the frequency of resonance in an open pipe changes with variations in the speed of sound due to temperature, and calculate the new resonant frequency when the speed is reduced by 3%.

End of Paper